

Question 1 Consider the following unconstrained problem

$$\begin{aligned} & \text{minimize} && f(x_1, x_2) \equiv x_1^2 - x_1x_2 + 2x_2^2 - 2x_1 + e^{x_1+x_2} \\ & \text{subject to} && (x_1, x_2) \in \mathbb{R}^2. \end{aligned}$$

- a) Write the first order necessary condition for the point $(\bar{x}_1, \bar{x}_2)$ to be a local minimum for this problem. (You do not need to solve the resulting equations.)
- b) Are the above conditions sufficient to guarantee that $(\bar{x}_1, \bar{x}_2)$ is a global minimum? If so, is this point a unique global minimum of $f(x_1, x_2)$?
- c) Is the point $(x_1, x_2) = (0, 0)$ a global minimum of $f(x_1, x_2)$? If not, identify a descent direction $d = (d_1, d_2)$ and find the minimum point of $f(x_1, x_2)$ along the line $\{\lambda d : \lambda \in \mathbb{R}\}$.

Question 2 State University has three professors who each teach four courses per year. Each year, four sections of Marketing, Finance, and Production must be offered. At least one section of each class must be offered during each semester (Fall and Spring). Each professor's time preference and preference for teaching various courses are given in the table below.

	<i>Professor 1</i>	<i>Professor 2</i>	<i>Professor 3</i>
<i>Fall Preference</i>	3	5	4
<i>Spring Preference</i>	4	3	4
<i>Marketing</i>	6	4	5
<i>Finance</i>	5	6	4
<i>Production</i>	4	5	6

The total satisfaction a professor earns teaching a class is the sum of the semester satisfaction and the course satisfaction. Thus, professor 1 derives a satisfaction of $3 + 6 = 9$ from teaching marketing during the fall semester. Formulate a minimum cost network flow problem that can be used to assign professors to courses so as to maximize the total satisfaction of the three professors. (**Hint:** Consider a network having 3 nodes associated with each professor, 3 nodes associated with each course and 6 nodes associated with each pair (semester, course).)

Question 3 The following network depicts a partially solved minimum cost network flow problem (MCNFP) with labels on nodes indicating net supply and those on arcs showing unit cost, upper bound, and current flow. (The lower bound on the flow through every arc is assumed to be zero.)

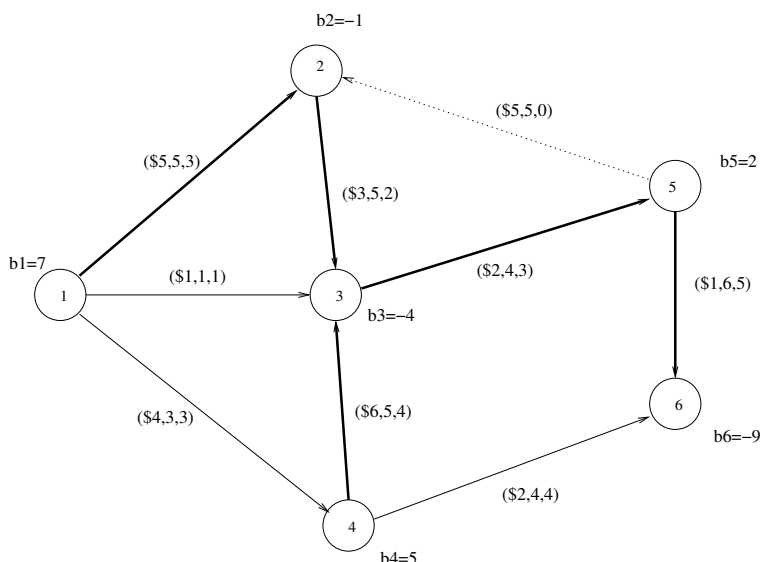


Figure 1: MCNFP Picture

- (7 points) Is the given solution an extreme point (or BFS) for the MCNFP?
- (7 points) Consider now performing one iteration of the minimum cost network simplex method. Find the values of the multipliers, that is, the u_i 's associated with each node.
- (8 points) Determine the reduced costs for each of the nonbasic arcs and verify that the current solution is not optimal. Which nonbasic arc flow should be modified so that the current flow can be improved?
- (8 points) Perform one pivot on the "best" available variable. Indicate the entering and leaving variables. Give the values of the new BFS obtained. By how much has the flow value improved?

Question 4 Consider the following LP problem in minimization form:

$$\begin{aligned}
 &\text{minimize} && w = 6y_1 - 4y_2 + 3y_3 - 6y_4 \\
 &\text{subject to} && y_1 - y_2 + y_3 + y_4 \geq 1 \\
 &&& -3y_1 + 2y_2 - y_3 + 2y_4 \leq -2 \\
 &&& y_1 \leq 0, \ y_2 \geq 0, \ y_3 \text{ urs}, \ y_4 \leq 0.
 \end{aligned}$$

- (a) Find the dual of this problem;
- (b) Graphically solve the dual problem;
- (c) Using the complementarity slackness theorem, find the optimal solutions of the above minimization problem. Is the optimal solution unique?